

Future Steps — Beyond the Public Evaluator

The public score is saturated (~0.92, hit@10 ~1.0). Marginal gains there are noise-fitting. This is the map of what is actually worth doing, why, and what it costs.

1. What the public evaluator rewards vs the four spec pillars

Judging is **Technical Execution 35% / Innovation 20% / Impact 20% / Feasibility 15% / Presentation 10%**. The automated `TechnicalScore` is explicitly "an objective input to the Technical Execution assessment... not a separate judging criterion" — so it is at most ~35% of one criterion.

Pillar (from the problem statement)	Value on the public evaluator
I — Dual-track routing	~0. The router only changes question wording; routing retrieval by intent was measured and dropped (dev flat, holdout -0.002).
I — LLM semantic ranking / vector similarity	~0. BM25 recall is already ~100%; the dense route never helped and has since been removed from the tree (change 20).
II — Incremental slots (accumulate constraints)	The whole game. 0.11 -> 0.78 came from this alone.
II — Slot <i>erasure</i> on override	Negative. The "erased" preference is still drawn from the target, so erasing it costs score; down-weighting / ignoring wins.
II — Proactive clarification / retrieval cutoff on over-generality	~0. The simulator always answers "anything else?", so <code>ask_attribute="other"</code> strictly dominates targeted questions.
III — Personalized context distillation (<code>user_profile</code>)	~0 to negative. Four independent attempts, all failed the holdout test.
III — Adaptive orchestration / runtime re-orchestration	Untested; the pipeline is deterministic and there is nothing to adapt to.

Why: a deterministic, templated, fully-cooperative simulator cannot reward dialogue intelligence — there is nothing to be clever about when the customer always answers on request. The `user_profile` is aggregate *past* taste, weakly linked to the specific target (only 26.5% of targets satisfy all their profile tags; for `intent_override` the profile aligns with neither side). So the pillars about *handling difficult customers* have no difficult customers to handle.

Why this is probably deliberate, not a mistake:

- 1 The number is <=35% of one of five criteria.
- 2 The pillars are listed under "Innovation Directions" — a pitch to human judges, not scored line-items.
- 3 The 800 private sessions "may paraphrase." If that simulator is harder (genuine decoys on override, sparser disclosure, paraphrased constraints), robust state handling, cue- (not template-) matching, and semantic retrieval become load-bearing insurance.
- 4 A demo of "anything else? ... anything else? ..." scoring 0.92 loses on Presentation and Innovation even while topping the objective input.

Strategic implication: the lean agent is the right *floor* — the feasibility baseline and the public number. Build the pillar features as **opt-in layers that do not regress the public score but demonstrably help on adversarial / paraphrased cases**. That is the Innovation + Impact pitch and the private-set hedge. Do not chase the last 0.005 on the leaderboard; 65% of judging is not there.

2. The "other, then InfoGain for more" hybrid policy

Right instinct for realism, **zero score impact on this simulator**. `customer_reply` has a hard cap of ~4 constraints per session. Once other has drained them, InfoGain asking about anything returns "no additional preference" — nothing left to extract. The premise ("users don't remember what they want until asked") is true for humans, false for

this simulator.

Build it as an opt-in that provably does not regress the public number: it is realism for the demo, and genuine insurance for the private set if that simulator discloses less on the first ask.

3. Improving the evaluator

You **cannot** change `evaluator/local_evaluator.py` for your reported score — the spec forbids it and it invalidates the number. Two legitimate moves:

- 1 **Build a separate realism harness** (extend `tools/hard_cases.py`). Synthesise sessions the official simulator cannot produce: paraphrased constraint disclosure, a *genuine decoy* on override, partial disclosure (customer volunteers 2, agent must dig for the other 2), contradictory constraints, a customer who says "I'm not sure" a lot. Measure against *this*, not just the public 200.
- 2 **Put the calibration critique in the write-up**. "Innovation & Problem Insight" (20%) is scored on how clearly the team framed the challenge. A section "what the public simulator can and can't measure, and how we tested beyond it" is exactly that criterion.

4. Remaining work, ranked

work	pillar / criterion	LLM needed?
Realism harness + testing against it	Problem Insight, Feasibility	No
Router that actually routes (narrow/confirm for buying, wide net for browsing) — currently only changes wording	I — Dual-Track	No
Proactive retrieval cutoff on over-generality — when the pool is huge and undifferentiated, ask a targeted question <i>before</i> showing a list	II — Proactive Guidance	Trigger: no. Question text: better with LLM
Explanation text — "these all have a stainless steel band and a date window" from the reranker's matched spans	listed Innovation Direction; Presentation	Optional (template works, LLM better)
LLM semantic rerank over the top ~20 when span coverage saturates (homogeneous clusters)	I — LLM Semantic Ranking	Yes
LLM-generated clarification questions from the pool's facet distribution	II — structured clarification	Yes
Adaptive orchestration — skip stages that cannot contribute; switch strategy on a stagnating session (diversify list / disambiguate)	III — Adaptive Orchestration	No
Paraphrase-robust constraint extraction (vs regex/cues) for the private set	II — state machine robustness	Yes

5. Does realism require an LLM?

Split it:

- ****Realism in what the agent says — varied natural clarification questions instead of the fixed QUESTION_TEXT table; explanations; handling messy / paraphrased customer input — yes, LLM-shaped.****
The repeated "Is there anything else that matters for this one?" is the single biggest realism gap and there is no non-LLM fix that isn't a bigger template table.

- ****Realism in what the agent *does* — retrieval, core ranking, state tracking, orchestration — no.**** BM25 recall is ~100%, exact-span matching is a strong ranking signal, and the state machine is deterministic logic.

Architecture: keep the offline lexical core as the guaranteed path and the score floor; add an LLM as **opt-in layers** on top — (1) question generation, (2) semantic rerank over top-K, (3) explanation text, (4) extraction fallback — each degrading to the current local behaviour if the model is unavailable. This keeps the "\$0, deterministic, offline" story *and* adds the "intelligent layer" story, and a network-restricted scoring run still gets 0.92.

6. Model options and what to expect

bge-small vs the BGE reranker

model	type	params	on-disk	can generate text?
bge-small-en-v1.5	bi-encoder embedding	33M	~130 MB	No — outputs a 384-dim vector
bge-reranker-base	cross-encoder	278M (~8x)	~1.1 GB fp32 / ~560 MB fp16	No — outputs one relevance score per (query, doc) pair
bge-reranker-large / -v2-m3	cross-encoder	~560M	~2.2 GB	No
bge-reranker-v2-gemma	LLM-based reranker	2.5B	~5 GB	Technically a decoder, but purpose-built to emit a score, not chat

Can BGE "do inference like an LLM"? The embedding and cross-encoder models cannot generate text at all — they are BERT-family encoders that output vectors or scores. Only the v2-gemma / v2-minicpm rerankers wrap a decoder LLM, and even those are tuned to score, not converse. For the clarification questions and explanations you need an actual generative model.

Speed (CPU, rough; GPU is 10-40x faster)

operation	latency
current agent, one turn	~17 ms
bge-small: encode one query	~2-8 ms
bge-reranker-base: score 20 candidates	~0.3-1.2 s per turn
Qwen2.5-0.5B (GGUF Q4): one ~25-token question	~1-2 s
Qwen2.5-1.5B: same	~2-4 s
Qwen2.5-3B / Llama-3.2-3B: same	~5-9 s
Qwen2.5-7B / Llama-3.1-8B: same	~12-25 s

So a small generative LLM on CPU is **100-1000x slower per call** than bge-small; the cross-encoder reranker sits in between (~10-100x bge-small, but 10-50x faster than a 3B LLM). A 3B model for question-gen on CPU adds ~5-15 s/turn -> ~1-2.5 min/session -> **hours for a full 200-session eval run**. On a T4-class GPU this collapses to sub-second and is a non-issue. Latency and cost are disclosed as feasibility metrics, not part of the score.

Expected score improvement

add	public	hard set	notes
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bge-small dense retrieval route	~0	~0	BM25 recall already ~100%. Built, measured flat-to-negative, and deleted in change 20 — reviving it means restoring <code>src/embed.py</code> and <code>tools/build_embeddings.py</code> from history, not flipping <code>dense_weight</code>
bge-reranker cross-encoder over top-20	~0 to +0.01	maybe +0.01-0.03 on <code>homogeneous_cluster / degenerate_card</code>	The winning rerank signal is <i>exact verbatim span match</i> — the disclosed constraints are literal catalog text. A semantic cross-encoder optimises for <i>meaning</i> , not string matching, so it may not beat span coverage on public and can even demote an exact-match target. Only clearly helps where span coverage saturates.
small LLM: clarification questions	0	0	The simulator ignores the English entirely. Pure Presentation / Innovation / demo value.
small LLM: extraction fallback	0	0	Helps only if the private simulator paraphrases; regex is fine on public.

Can a small LLM produce good responses?

For **narrow, well-scoped rewriting/formatting** — "turn these facet options into a natural question", "explain why these 3 products match given these matched phrases" — **yes**, a 1.5-3B instruct model (Qwen2.5, Llama-3.2, Phi-3.5-mini) is acceptable, provided the actual content is supplied in the prompt and the output is validated with a template fallback. 0.5B is too weak (unreliable instruction following). For open-ended dialogue or nuanced reasoning a small model is mediocre and hallucination-prone — do not ask it to *decide* anything, only to *phrase* things whose content the deterministic pipeline already produced. 7-8B is where quality gets genuinely good, at ~4-5 GB and ~2-4x the latency.

Bottom line on models: an offline stack of `bge-small` (retrieval, already cached) + `bge-reranker-base` (rerank tie-breaker over top-20) + a ~3B instruct model (questions + explanations, template fallback) keeps everything local, adds ~1-2 s/turn on GPU, and buys demo realism plus a modest hard-set gain — not a public-score jump.